

Converting between Sequent Systems and Truth Trees

Andrew Nazareth and Andrii Bonenberger

March 31, 2026

Sequent System Rules (Propositional Calculus)

The sequent system we are using has the following rules of operation. Γ and Δ are sets (multisets) and A and B are atomics.

Operational Rules

Negation

$$\frac{\Gamma \Rightarrow \Delta, A}{\neg A, \Gamma \Rightarrow \Delta} \neg L \qquad \frac{A, \Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, \neg A} \neg R$$

Conjunction

$$\frac{A, B, \Gamma \Rightarrow \Delta}{A \wedge B, \Gamma \Rightarrow \Delta} \wedge L \qquad \frac{\Gamma \Rightarrow \Delta, A \quad \Gamma \Rightarrow \Delta, B}{\Gamma \Rightarrow \Delta, A \wedge B} \wedge R$$

Disjunction

$$\frac{A, \Gamma \Rightarrow \Delta \quad B, \Gamma \Rightarrow \Delta}{A \vee B, \Gamma \Rightarrow \Delta} \vee L \qquad \frac{\Gamma \Rightarrow \Delta, A, B}{\Gamma \Rightarrow \Delta, A \vee B} \vee R$$

Implication

$$\frac{\Gamma \Rightarrow \Delta, A \quad B, \Pi \Rightarrow \Lambda}{A \rightarrow B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} \rightarrow L \qquad \frac{A, \Gamma \Rightarrow \Delta, B}{\Gamma \Rightarrow \Delta, A \rightarrow B} \rightarrow R$$

Structural Rules

Weakening

$$\frac{\Gamma \Rightarrow \Delta}{A, \Gamma \Rightarrow \Delta} WL \qquad \frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, A} WR$$

Contraction

$$\frac{A, A, \Gamma \Rightarrow \Delta}{A, \Gamma \Rightarrow \Delta} CL \qquad \frac{\Gamma \Rightarrow \Delta, A, A}{\Gamma \Rightarrow \Delta, A} CR$$

Exchange

$$\frac{\Gamma, A, B, \Pi \Rightarrow \Delta}{\Gamma, B, A, \Pi \Rightarrow \Delta} XL \qquad \frac{\Gamma \Rightarrow \Delta, A, B, \Lambda}{\Gamma \Rightarrow \Delta, B, A, \Lambda} XR$$

File Layout

The sequent system will be formatted as a json with a list of nodes containing their node ID, antecedent/succedent, the node ids used to get there and the rule used to get there, going from goal upwards. For example, the following sequent:

$$\frac{\frac{\frac{A \Rightarrow A}{B, A \Rightarrow A} WL}{A, B \Rightarrow A} XL \quad \frac{\frac{B \Rightarrow B}{A, B \Rightarrow B} WL}{A, B \Rightarrow A \wedge B} \wedge R$$

would be represented as: !!! TODO !!!

Truth Tree Rules (Propositional Calculus)

Decomposition Rules

Closure Rules

File Layout

The truth tree will be formatted into a json-like form for use in Willow.

Putting them together